

Fig. 6: An actual-size, single-side PCB for the RFID-based security system

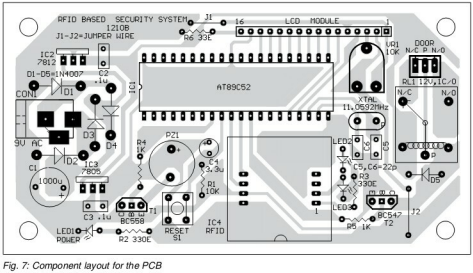


Fig. 7: Component layout for the PCB

greater range than passive tags.

Semi-passive tags have their own power source that powers only the microchip. These have no transmitter. They rely on altering the RF field from the transceiver to transmit their data.

There are three ways for data encoding into tags:

1. Read-only tags contain data, which is pre-written onto them by the tag manufacturer or distributor.

2. Write-once tags enable a user to write data to the tag one time in production or distribution processes.

3. Full read-write tags allow new data to be written to the tag as needed and later other data can be rewritten over the original data.

RF transceiver. The RF transceiver is the source of the RF energy used to activate and power the passive RFID tags. It may be enclosed in the same cabinet as the reader or it may be a separate piece of equipment. When provided as a separate piece of equipment, the transceiver is commonly referred to as an RF module. The RF transceiver controls and modulates the radio frequencies that the antenna transmits and receives. The transceiver filters and amplifies the back-scatter signal from a passive RFID tag.

Circuit description

Fig. 5 shows the circuit of the RFID-based security system. The compact circuitry is built around Atmel AT89C52 microcontroller. The AT89C52 is a low-power, high-performance CMOS 8-bit microcomputer with 8 kB of Flash programmable and erasable read-only memory (PEROM). It has 256 bytes of RAM, 32 input/output (I/O) lines, three 16-bit timers/counters, a six-vector two-level interrupt architecture, a full-duplex serial port, an on-chip oscillator and clock circuitry. The system clock also plays a significant role in operation of the microcontroller.

An 11.0592MHz quartz crystal connected to pins 18 and 19 provides basic clock to the microcontroller. Power-on reset is provided by the combination of electrolytic capacitor C4 and resistor R1. Switch S1 is used for manual reset. Port pins P2.0 through P2.7 of the microcontroller are connected to data port pins D0 through D7 of the LCD, respectively. Port pins P3.7 and P3.6 of the microcontroller are connected to register-select (RS) and enable (E) pins of the LCD, respectively. Read/write (R/$\overline{W}$) pin of the LCD is grounded to enable for write operation.

All the data is sent to the LCD in ASCII format for display. Only the commands are sent in hex form. Register-select (RS) signal is used to distinguish between data (RS=1) and command (RS=0). Preset VR1 is used to control the contrast of the LCD. Resistor R6 limits the current through the backlight of the LCD. Port pins P3.0 (R_{XD}) and P3.1 (T_{XD}) of the microcontroller are used to interface with the RFID reader.

When an authorised person having the tag enters the RF field generated by the RFID reader, RF signal is generated by the RFID reader to transmit energy to the tag and retrieve data from the tag. Then the RFID reader communicates through R_{XD} and T_{XD} pins of the microcontroller for further processing.